

Graphical Abstract

An Explainable Convolutional Neural Network Framework for Personal Protective Equipment Compliance Detection on Construction Sites

Pavan Pitta

PPE Compliance Detection from Construction-Site Imagery



Best model: Compact CNN — 74.5% test accuracy, 0.90 macro-AUC

Highlights

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- Four-class worker-level PPE compliance recognition on a public construction-site image dataset.
- Person-centred cropping turns scene-level detection labels into a tractable classification task.
- A compact custom CNN reaches 74.5 percent test accuracy with only 0.42 million parameters.
- Partial-compliance classes such as hard-hat-only are the hardest to recognise reliably.
- Grad-CAM confirms the model attends to the head and torso where PPE is worn.

An Explainable Convolutional Neural Network Framework for Personal Protective Equipment Compliance Detection on Construction Sites

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Abstract

Personal protective equipment (PPE) such as hard hats and high-visibility safety vests is a primary line of defence against injury on construction sites, yet manual monitoring of compliance across large and crowded worksites is slow, error-prone, and impossible to sustain in real time. Most published PPE studies frame the problem as bounding-box object detection and report only aggregate detection scores, rarely pairing an image-level compliance decision with per-class reliability analysis, a controlled architecture comparison, and interpretability in a single evaluation. This study reformulates worker-level PPE monitoring as a four-class image-classification problem, deriving compliance labels (both items, hard hat only, vest only, or no PPE) from a public construction-site dataset and extracting 7,462 person-centred crops so that the equipment of interest is actually visible in each input. Two convolutional neural networks trained from scratch under identical conditions are compared: a compact network with roughly 0.42 million parameters and a deeper strided network with roughly 2.5 million parameters, both using softened class weighting to counter the strong imbalance toward the no-PPE class. On a held-out test set of 1,120 crops the compact network reached an accuracy of 74.5 percent, a macro F1-score of 0.686, and a macro one-vs-rest AUC of 0.902, slightly outperforming the deeper network at 70.9 percent accuracy and 0.888 AUC. A per-class analysis showed that the fully compliant and no-PPE classes are recognised reliably, while the partial-compliance classes, especially hard-hat-only, are the hardest because a single small item must be distinguished under occlusion, distance, and viewpoint variation. Grad-CAM analysis confirmed that the compact model attends to the head and torso regions where PPE is worn rather than to the background. The

contribution is a reproducible, explainable, and efficiency-aware PPE compliance classification study that establishes an honest from-scratch baseline and characterises exactly where the models succeed and fail.

Keywords: PPE compliance detection, Construction safety, Convolutional neural networks, Explainable AI, Grad-CAM, Image classification

1. Introduction

1.1. Application Background and Practical Importance

Construction remains one of the most hazardous industrial sectors, consistently accounting for a disproportionate share of occupational injuries and fatalities worldwide. A large fraction of these incidents involve head trauma and struck-by events that correct use of personal protective equipment, principally hard hats and high-visibility safety vests, is designed to prevent. Regulators such as the Occupational Safety and Health Administration mandate PPE use, and non-compliance exposes workers to serious harm and exposes employers to substantial fines and legal liability. Despite these mandates, ensuring that every worker wears the required equipment at every moment is difficult on active sites that may span many zones and hundreds of workers. Human supervisors cannot observe every worker continuously, and reviewing hours of closed-circuit television footage after an incident is far too slow to prevent it. The scale and dispersion of modern worksites therefore create a persistent gap between the existence of safety rules and their moment-to-moment enforcement. Automated visual monitoring that flags non-compliance in real time offers a scalable way to close this gap and to shift safety management from reactive to preventive. This motivates computer-vision systems that can decide, from a single camera frame, whether a worker is properly equipped [1, 2].

1.2. Machine Learning and Computer Vision Context

Convolutional neural networks have become the dominant tool for image understanding because they learn hierarchical visual features directly from pixels rather than relying on hand-crafted descriptors [3, 4]. Early convolutional layers respond to edges and colours, while deeper layers compose these into parts and object-level concepts, which is exactly the kind of representation needed to recognise a helmet or a vest. For safety monitoring, this capacity to learn appearance under varied lighting, pose, and background

is essential, since construction imagery is visually cluttered and highly variable. Most existing PPE work adopts object-detection frameworks such as the YOLO family that localise each equipment item with a bounding box [5, 6, 7]. Detection is powerful but produces box-level outputs that still require a decision rule to determine whether a given worker is compliant. An alternative and complementary formulation treats the worker region itself as the unit of analysis and classifies its overall compliance state directly. This classification view is attractive for teaching and for lightweight deployment because it yields a single interpretable label per worker and trains stably on modest hardware. The present study adopts this worker-level classification formulation and studies it rigorously with custom convolutional networks.

1.3. Custom Architectures and the Depth Question for the Selected Problem

A central design question for any classification task is how much model capacity the problem actually needs. Very deep networks such as residual architectures can extract extremely rich features but carry tens of millions of parameters and high computational cost [8]. Lightweight architectures such as MobileNetV2 trade a little accuracy for dramatic reductions in parameters and inference time, which matters for edge deployment on a construction site [9]. Pre-trained weights from large datasets such as ImageNet usually give transfer-learning models a strong head start, but they are not always available in restricted or offline training environments. In such settings it is important to understand how far a network trained purely from scratch on the available data can go, and whether adding depth helps or simply invites overfitting on a modest dataset. This study therefore compares two custom convolutional networks of different depth, both trained from random initialisation under identical conditions, isolating the effect of capacity on this specific task. The comparison provides an honest baseline that is directly reproducible without any external pre-trained weights. It also clarifies whether, for worker-level PPE recognition on a few thousand images, a small efficient model is sufficient.

1.4. Explainable AI and Trustworthiness

A safety-critical system that silently outputs a label is difficult to trust, because operators cannot tell whether a decision rests on the right visual evidence. Explainable AI methods address this by exposing which parts of an image drove a prediction [10]. Gradient-weighted Class Activation Mapping,

65 or Grad-CAM, produces a heatmap highlighting the image regions most re-
66 sponsible for a class score [11]. For PPE recognition, a trustworthy model
67 should attend to the head region when reasoning about hard hats and to
68 the torso when reasoning about vests, rather than to background scaffolding
69 or vehicles. Grad-CAM lets us verify this expectation and diagnose failure
70 cases where the model latches onto misleading cues such as bright background
71 colours. Pairing quantitative metrics with these visual explanations turns a
72 black-box classifier into an auditable one. This study integrates Grad-CAM
73 directly into the evaluation so that correct and incorrect predictions can be
74 inspected side by side. Interpretability is treated here as a first-class result
75 rather than an afterthought.

76 1.5. Research Gap

77 Although PPE detection is an active research area, several gaps persist
78 in the way results are reported and validated. First, the large majority of
79 studies report only aggregate detection or classification accuracy and omit
80 per-class reliability, which hides the fact that partial-compliance cases are
81 far harder than the easy extremes [12, 13]. Second, controlled architecture
82 comparisons under identical data splits, augmentation, and training sched-
83 ules are rare, so it is often unclear whether a reported gain comes from the
84 model or from incidental differences in the pipeline. Third, many works de-
85 pend on pre-trained backbones and do not report how a from-scratch model
86 performs, leaving the true difficulty of the raw task unmeasured. Fourth, in-
87 terpretability is frequently absent, so readers cannot judge whether a model
88 relies on genuinely PPE-informative evidence. Fifth, the common object-
89 detection framing does not directly yield a clean worker-level compliance
90 decision that a supervisor can act on. This study addresses these gaps to-
91 gether by adopting a worker-level classification formulation, comparing two
92 from-scratch architectures under strictly identical conditions, reporting full
93 per-class metrics, and integrating Grad-CAM interpretation. The result is
94 a transparent and honest characterisation of the task rather than a single
95 headline number.

96 1.6. Problem Statement

97 The problem addressed in this study is the automatic classification of a
98 worker’s PPE compliance state from a single cropped image region. Given
99 an image containing a person on a construction site, the system must assign
100 one of four mutually exclusive labels: the worker wears both a hard hat and a

101 safety vest; the worker wears a hard hat only; the worker wears a safety vest
 102 only; or the worker wears neither item. This is a supervised multi-class image-
 103 classification task with four classes and a strong class imbalance driven by the
 104 frequency of non-compliant workers in the source data. The inputs are RGB
 105 person crops of variable original size, resized to a fixed resolution, and the
 106 outputs are calibrated class probabilities. The task is challenging because the
 107 discriminative equipment can be small, partially occluded, seen from behind,
 108 or confused with similarly coloured background objects. A successful system
 109 must maintain balanced accuracy across all four classes, not merely on the
 110 easy majority class. The study measures this with per-class precision, recall,
 111 and F1-score in addition to overall accuracy and one-vs-rest AUC.

112 *1.7. Aim and Objectives*

113 The aim of this study is to develop and rigorously evaluate an explain-
 114 able convolutional neural network system for worker-level PPE compliance
 115 classification, and to determine how model depth affects performance when
 116 training from scratch. The first objective is to construct a clean four-class
 117 classification dataset from a public construction-site source by extracting
 118 person-centred crops and deriving compliance labels from the underlying an-
 119 notations. The second objective is to design a compact custom convolutional
 120 network as an efficient baseline and a deeper custom network as a higher-
 121 capacity comparator. The third objective is to train both networks under
 122 identical preprocessing, augmentation, class-weighting, and optimisation set-
 123 tings so that the comparison isolates architecture. The fourth objective is to
 124 evaluate both models with a comprehensive metric suite including per-class
 125 scores, confusion matrices, and ROC analysis. The fifth objective is to apply
 126 Grad-CAM to interpret the models and to inspect representative correct and
 127 incorrect predictions. The final objective is to report limitations honestly
 128 and to outline concrete directions for future improvement.

129 *1.8. Research Questions*

130 This study is organised around five research questions. RQ1 asks whether
 131 a convolutional network trained from scratch on roughly seven thousand
 132 person crops can achieve a useful and balanced classification of the four PPE
 133 compliance states. RQ2 asks how a deeper custom architecture compares to
 134 a compact one, in terms of accuracy, macro F1-score, and AUC, when both
 135 are trained under identical conditions without pre-trained weights. RQ3 asks
 136 which data-preparation and augmentation choices most effectively counter

137 overfitting and class imbalance on a dataset of this size. RQ4 asks which
138 PPE compliance class is hardest to classify and what visual factors, revealed
139 by Grad-CAM and error analysis, explain the confusions. RQ5 asks whether
140 a compact and computationally cheap model is sufficient for this task, which
141 would make real-time edge deployment feasible.

142 *1.9. Contributions and Novelty*

143 The contributions of this study are fivefold. First, it reformulates construction-
144 site PPE monitoring as a clean four-class worker-level classification problem
145 and documents a reproducible procedure for deriving it from a public object-
146 detection dataset through person-centred cropping. Second, it provides a
147 controlled comparison of two from-scratch convolutional architectures of dif-
148 fering depth under strictly identical conditions, establishing an honest base-
149 line free of pre-trained weights. Third, it reports a complete per-class reliabil-
150 ity analysis rather than a single accuracy figure, exposing exactly where the
151 models are strong and weak. Fourth, it integrates Grad-CAM interpretation
152 to verify that predictions rest on PPE-informative regions and to diagnose
153 failure modes. Fifth, it offers a candid discussion of limitations and future
154 work, including the value of pre-trained backbones and higher-resolution in-
155 puts, that situates the results within the broader literature. Together these
156 contributions prioritise transparency and reproducibility over headline num-
157 bers.

158 *1.10. Report Organization*

159 The remainder of this report is organised as follows. Section 2 reviews
160 related work on vision-based construction safety, CNN architectures, data
161 augmentation, evaluation practice, and explainability, and summarises the
162 research gap in a comparative matrix. Section 3 describes the methodology,
163 including the dataset, label derivation, preprocessing, the two network archi-
164 tectures, the training procedure, the evaluation framework, and the Grad-
165 CAM methodology. Section 4 presents the experimental setup and the full
166 quantitative and qualitative results. Section 5 discusses the findings in rela-
167 tion to the research questions and the literature, and states the limitations
168 and future work. Section 6 concludes the report. All figures and tables are
169 referred to from the main text, and the code and dataset preparation steps
170 are released for reproducibility.

171 2. Literature Review

172 2.1. Vision-Based Safety Monitoring in Construction

173 Computer vision for construction safety has grown rapidly as cameras
174 have become ubiquitous on worksites and deep learning has matured [2].
175 Early influential work detected the absence of hard hats from far-field surveil-
176 lance video, demonstrating that convolutional models could flag a specific un-
177 safe condition automatically [14]. Subsequent studies broadened the scope
178 from single-item detection to multi-item PPE recognition covering helmets,
179 vests, and other gear [15]. The seminal work of Nath and colleagues framed
180 real-time PPE detection as a deep-learning problem and provided a widely
181 cited reference pipeline [1]. These studies established that vision-based mon-
182 itoring is technically feasible and that its main challenges are appearance
183 variation, occlusion, and the small apparent size of equipment at distance.
184 They also showed that domain-specific data and careful labelling matter as
185 much as the choice of network. The present study builds on this line by fo-
186 cusing on a worker-level compliance decision rather than raw item detection.

187 2.2. CNN and Detection Methods for PPE

188 A large share of recent PPE literature adopts object detectors, particu-
189 larly the YOLO family, for their speed and accuracy [5, 6, 16]. These systems
190 localise each PPE item and often infer compliance by spatial reasoning about
191 which items overlap which person. Detection-based systems achieve strong
192 mean average precision but require post-processing rules to convert boxes
193 into a per-worker compliance verdict, and these rules are rarely evaluated in
194 isolation [17]. Classification-based approaches, by contrast, assign a label to
195 a person region directly and are simpler to interpret and deploy [18]. Several
196 works combine detection to localise workers with a downstream classifier that
197 judges their equipment, which is conceptually close to the person-cropping
198 strategy adopted here [19]. Lightweight and edge-oriented systems have also
199 been proposed to meet real-time constraints on-site [20]. The literature thus
200 supports a two-stage view in which localisation and compliance classification
201 are distinct sub-problems, and this study concentrates on the classification
202 stage.

203 2.3. Custom versus Pre-Trained Architectures

204 The choice between training from scratch and transfer learning is a recur-
205 ring theme in applied deep learning [21]. Transfer learning from ImageNet-
206 pre-trained backbones such as ResNet and MobileNet usually accelerates

207 convergence and improves accuracy, especially when data are limited [8, 9].
 208 However, pre-trained weights are not always accessible, for example in offline
 209 or access-restricted training environments, which makes from-scratch perfor-
 210 mance an important and honestly reported baseline. Efficiency-focused ar-
 211 chitectures like EfficientNet and MobileNetV2 show that careful scaling can
 212 preserve accuracy at a fraction of the parameters [22, 9]. Very deep models
 213 can overfit small datasets, so greater depth is not automatically better when
 214 only a few thousand images are available. This tension between capacity and
 215 generalisation is precisely what the controlled depth comparison in this study
 216 probes. By training both a compact and a deeper network from scratch, the
 217 study measures whether added depth pays off on this modest dataset.

218 *2.4. Data Preprocessing and Augmentation*

219 Data augmentation is a standard and effective defence against overfitting
 220 when training data are scarce [23]. Common transformations include hori-
 221 zontal flipping, small rotations, zoom, and brightness perturbation, each of
 222 which reflects plausible real-world variation on a worksite. Studies specific to
 223 construction imagery have tuned augmentation hyperparameters and shown
 224 measurable gains in classification robustness [24]. Class imbalance is another
 225 pervasive issue, since non-compliant or majority classes often dominate real
 226 datasets, and it is typically addressed with class weighting or resampling.
 227 Normalisation of pixel intensities and consistent input resizing are also es-
 228 sential for stable convolutional training [25]. Regularisation techniques such
 229 as dropout further reduce overfitting in the fully connected head [26]. This
 230 study applies a conservative combination of flipping, rotation, and zoom
 231 augmentation together with softened class weighting, following the balance
 232 recommended in this literature.

233 *2.5. Evaluation and Reliability*

234 Sound evaluation practice requires more than a single accuracy number,
 235 particularly under class imbalance [12]. Per-class precision, recall, and F1-
 236 score reveal whether a model is merely exploiting the majority class or is gen-
 237 uinely balanced. Confusion matrices expose the specific structure of errors,
 238 such as systematic confusion between visually similar classes. One-vs-rest
 239 ROC analysis and the area under the curve summarise the discriminative
 240 power of the model independently of a single decision threshold. Macro-
 241 averaged metrics weight all classes equally and are therefore the fairest sum-
 242 mary when minority classes matter for safety. Reporting training and val-

243 idation curves also allows overfitting to be diagnosed directly. This study
244 adopts all of these practices to give a complete and honest picture of model
245 behaviour.

246 *2.6. Explainable AI in Safety Applications*

247 Interpretability is increasingly regarded as a requirement rather than a
248 luxury for safety-critical vision systems [10]. Grad-CAM has become the de
249 facto tool for visualising which image regions drive a convolutional model’s
250 decision [11]. In safety monitoring, such explanations let auditors confirm
251 that a helmet decision depends on the head region and not on incidental
252 background cues. Explanations are equally valuable for debugging, since
253 they reveal when a model succeeds for the wrong reasons or fails because
254 of misleading context. Relatively few PPE studies report interpretability
255 alongside their metrics, which is one of the gaps this study targets. By
256 pairing Grad-CAM with the quantitative evaluation, the study makes the
257 models auditable. This aligns with the broader movement toward responsible
258 and transparent AI in high-stakes domains.

259 *2.7. Comparative Literature Review Matrix*

260 Table 1 summarises representative related studies against the dimensions
261 that matter for this work, namely the task framing, the model family, whether
262 a controlled multi-model comparison is performed, whether class balancing
263 is used, whether interpretability is reported, and whether a from-scratch
264 baseline is provided. The matrix shows that most prior studies emphasise
265 detection accuracy and rarely combine a controlled architecture comparison,
266 full per-class reliability, and interpretability in one place. It also shows that
267 from-scratch baselines are underreported, which is exactly the contribution
268 positioned in the final row for the present study.

269 *2.8. Synthesis of Literature and Identified Gap*

270 Taken together, the literature demonstrates that vision-based PPE mon-
271 itoring is feasible and valuable, that detection dominates the task framing,
272 and that transfer learning is the usual route to high accuracy. It also re-
273 veals consistent gaps: controlled from-scratch architecture comparisons are
274 rare, per-class reliability is often omitted, and interpretability is seldom inte-
275 grated. The worker-level classification framing adopted here directly yields
276 the actionable compliance verdict that detection pipelines must reconstruct
277 with additional rules. By comparing two from-scratch architectures under

Table 1: Comparative literature review matrix summarising task framing, model family, and evaluation practices in closely related PPE and safety-recognition studies, with the final row describing the present study. Abbreviations: Det = detection; Cls = classification; TL = transfer learning; MC = multiple-model comparison; CB = class balancing; XAI = explainability (Grad-CAM); FS = from-scratch baseline. ✓ = included, ✗ = not included, ~ = partial.

Study	Framing	Model family	MC	CB	XAI	FS
Fang et al. [14]	Det	Faster R-CNN	✗	~	✗	✗
Nath et al. [1]	Det	YOLO	~	✗	✗	✗
Delhi et al. [15]	Det	YOLOv3	✗	✗	✗	✗
Wang et al. [6]	Det	YOLO variants	~	✗	✗	✗
Ferdous & Ahsan [5]	Det	YOLO	~	✗	✗	✗
Isailovic et al. [17]	Det+Cls	CNN detectors	~	~	✗	✗
Gallo et al. [19]	Det+Cls	Deep CNN	✗	~	✗	✗
Ahmed & Rahman [18]	Cls	Deep CNN	~	~	✗	✗
This study	Cls	Custom CNNs	✓	✓	✓	✓

identical conditions, reporting complete per-class metrics, and adding Grad-CAM, this study fills the identified gap. It deliberately trades a headline accuracy number for transparency, reproducibility, and an honest account of task difficulty. This positioning motivates the methodology described next.

3. Methodology

3.1. Overview of the Pipeline

The methodology follows a seven-stage pipeline that transforms raw construction-site imagery into an interpretable four-class compliance classifier, as illustrated in Figure 1. The first stage acquires the public construction-site safety dataset and inspects its annotation format. The second stage extracts person-centred crops from each image using the underlying bounding boxes, so that the equipment of interest occupies a meaningful fraction of the input. The third stage derives a single four-class compliance label for each crop from the co-located PPE annotations. The fourth stage preprocesses the crops by resizing and normalisation and splits them into training, validation, and test partitions with stratification. The fifth stage trains two custom convolutional networks of differing depth under identical augmentation, class-weighting, and optimisation settings. The sixth stage evaluates both models with a comprehensive metric suite, and the seventh stage applies

297 Grad-CAM to interpret their decisions. Each stage is deterministic given a
 298 fixed random seed, which makes the whole pipeline reproducible.

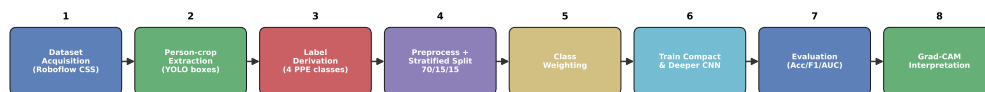


Figure 1: End-to-end methodology pipeline for worker-level PPE compliance classification. Raw construction-site images are converted into person-centred crops, labelled into four compliance classes, preprocessed and split with stratification, used to train a compact and a deeper custom CNN under identical conditions, evaluated with per-class metrics and ROC analysis, and finally interpreted with Grad-CAM.

299 3.2. Dataset Source and Description

300 The study uses the publicly available Construction Site Safety Image
 301 Dataset distributed through Roboflow Universe, which contains construction-
 302 scene photographs annotated for ten object categories in YOLO format [27].
 303 The relevant categories for compliance reasoning are hard hat, no-hard-hat,
 304 safety vest, no-safety-vest, and person, while the remaining categories such
 305 as machinery and vehicle provide scene context but are not used for labelling.
 306 The original annotations are bounding boxes, one per object instance, ex-
 307 pressed as normalised centre coordinates and dimensions. Because the raw
 308 task is scene-level object detection, the dataset is not directly a classification
 309 dataset, and a derivation step is required to obtain worker-level labels. The
 310 scenes are visually diverse, spanning different sites, camera distances, view-
 311 points, lighting conditions, and degrees of occlusion, which makes the derived
 312 classification task realistically difficult. The imagery is genuine surveillance-
 313 style construction footage rather than staged photographs, so it reflects the
 314 conditions a deployed system would face. Table 2 summarises the key at-
 315 tributes of the dataset as used in this study.

316 3.3. Person-Crop Extraction and Label Derivation

317 The core preprocessing idea is that scene-level images place the discrimi-
 318 native equipment in a small, variable region, which makes direct scene classi-
 319 fication unreliable, whereas cropping to each person concentrates the useful

Table 2: Summary of the dataset as adapted for worker-level PPE compliance classification. Person-centred crops are extracted from the source construction-site images and labelled into four mutually exclusive compliance classes.

Attribute	Details
Source dataset	Construction Site Safety Image Dataset (Roboflow Universe)
Original task	Object detection, 10 categories, YOLO format
Derived task	4-class worker-level PPE compliance classification
Unit of analysis	Person-centred crop (one per detected person)
Total labelled crops	7,462
Number of classes	4 (PPE_Compliant, HardHat_Only, Vest_Only, NO_PPE)
Image format	RGB (JPEG), variable original size
Model input size	$96 \times 96 \times 3$, pixel values scaled to $[0, 1]$
Split strategy	Stratified 70% train / 15% validation / 15% test

signal. For every image, each person bounding box is used to crop the corresponding region, which is then resized to the fixed model input size. The compliance label for a crop is derived by checking which PPE-related bounding boxes fall within that person’s box, using a centre-containment test. If a hard-hat box and a safety-vest box are associated with the person, the crop is labelled PPE_Compliant; if only a hard hat is present, it is HardHat_Only; if only a vest is present, it is Vest_Only; and if only negative markers such as no-hard-hat or no-safety-vest are present, it is NO_PPE. Crops that are too small in width or height are discarded to avoid degenerate inputs, and ambiguous cases that do not map cleanly to one of the four classes are excluded. This procedure yields 7,462 clean, labelled person crops from the source images. The derivation is fully automatic and deterministic, so the resulting dataset can be regenerated exactly from the public source. Figure 2 shows representative crops for each of the four classes, illustrating both the visual signal the model can exploit and the variability it must tolerate.

3.4. Class Distribution and Imbalance

The derived dataset is imbalanced, reflecting the real frequency of each compliance state in the source imagery. The NO_PPE class is the largest with 3,596 crops, followed by PPE_Compliant with 1,657, HardHat_Only with 1,227, and Vest_Only with 982. This imbalance is important because a naive classifier could achieve deceptively high accuracy simply by favouring the majority class while failing on the safety-critical minority classes. Fig-



Figure 2: Representative person-centred crops for each of the four PPE compliance classes. Each row shows a different class (top to bottom: PPE_Compliant, HardHat_Only, Vest_Only, NO_PPE). The crops illustrate the discriminative cues (helmet on the head, vest on the torso) as well as the occlusion, viewpoint, and lighting variation that make the task challenging.

ure 3 visualises the class distribution across the full set of crops. To prevent
the majority class from dominating training, softened class weights inversely
proportional to class frequency are applied to the loss, which raises the effective
importance of the minority classes without the instability of aggressive

346 resampling. Stratified splitting further ensures that each of the training, val-
 347 idation, and test partitions preserves the same class proportions. Together
 348 these measures make both the training signal and the evaluation fair across
 349 all four classes.

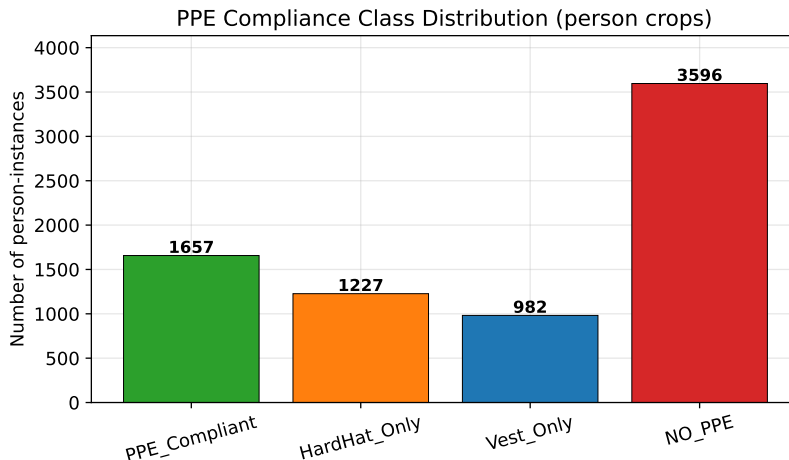


Figure 3: Distribution of person-centred crops across the four PPE compliance classes. The dataset is imbalanced toward the NO_PPE class, which motivates the softened class-weighting scheme and the stratified train/validation/test split used in this study.

350 3.5. Preprocessing, Splitting, and Augmentation

351 All crops are resized to 96×96 pixels in RGB and their pixel intensities
 352 are scaled to the unit interval, which stabilises convolutional training [25].
 353 The dataset is partitioned into 70 percent training, 15 percent validation,
 354 and 15 percent test using stratified sampling with a fixed random seed, giv-
 355 ing 5,223 training, 1,119 validation, and 1,120 test crops. The validation
 356 partition is used for early stopping, learning-rate scheduling, and checkpoint
 357 selection, while the test partition is held out entirely for the final reported
 358 metrics. Data augmentation is applied on the training stream only, and com-
 359 prises horizontal flipping, small random rotation of up to roughly five degrees,
 360 and random zoom of up to ten percent. These conservative transformations
 361 reflect realistic variation in worker pose and camera framing without dis-
 362 torting the equipment beyond recognition [23, 24]. Brightness and heavier
 363 geometric perturbations were deliberately avoided after preliminary experi-
 364 ments showed they slowed convergence on this small dataset. The identical

preprocessing and augmentation pipeline is used for both networks so that any performance difference is attributable to architecture alone.

3.6. Compact Custom CNN Architecture

The first model is a compact convolutional network designed as an efficient baseline, shown schematically in Figure 4. It consists of four convolutional blocks with 32, 64, 128, and 256 filters respectively, each block applying a 3×3 convolution followed by batch normalisation and a ReLU activation, and then 2×2 max pooling that halves the spatial resolution [25]. After the convolutional stack, a global average pooling layer collapses the feature maps into a compact feature vector, which avoids the large parameter count of a flattened dense connection. A single fully connected layer of 128 units with ReLU activation and dropout at a rate of 0.3 forms the classification head, followed by a four-unit softmax output [26]. The global-average-pooling design keeps the model small, at roughly 0.42 million trainable parameters, which is attractive for edge deployment. Batch normalisation after each convolution accelerates and stabilises training, while dropout in the head limits overfitting on the modest dataset. This architecture embodies the hypothesis that a small, well-regularised network is sufficient for worker-level PPE recognition.

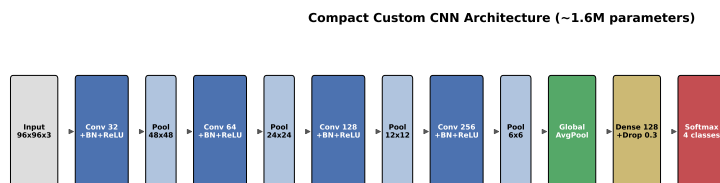


Figure 4: Architecture of the compact custom convolutional network. Four convolutional blocks with increasing filter depth (32, 64, 128, 256), each with batch normalisation, ReLU, and max pooling, are followed by global average pooling, a 128-unit dense layer with dropout, and a four-way softmax output. The design totals approximately 0.42 million trainable parameters.

3.7. Deeper Custom CNN Architecture

The second model is a deeper convolutional network intended to test whether additional capacity improves performance under identical condi-

tions. It replaces single convolutions with paired convolutions at each resolution and uses strided convolutions to perform spatial downsampling, which is more expressive than plain pooling while remaining efficient. The network progresses through filter depths of 32, 64, 128, and 256 with strided reduction from 96×96 down to 6×6 , and a final 512-filter convolutional block captures high-level abstractions. As in the compact model, global average pooling feeds the classification head, but the head is larger, with a 256-unit dense layer followed by a 128-unit dense layer, each with dropout, before the four-way softmax. This design totals roughly 2.5 million trainable parameters, about six times the compact model, giving it substantially greater representational capacity. Batch normalisation is applied throughout to keep the deeper stack trainable from random initialisation. The two architectures differ only in depth and head size, so the comparison cleanly isolates the effect of capacity.

3.8. Training Procedure

Both networks are trained from random initialisation, without any pre-trained weights, using the Adam optimiser and the sparse categorical cross-entropy loss [28]. Softened class weights inversely proportional to class frequency are passed to the loss so that minority classes contribute proportionally more to the gradient. Training uses a batch size of 32 and runs for up to 40 to 45 epochs, with early stopping monitoring validation accuracy and restoring the best weights to prevent overfitting. A ReduceLROnPlateau schedule lowers the learning rate when validation accuracy stalls, which allows the models to refine once coarse features are learned. Model checkpointing saves the best-performing weights on the validation set, and these checkpoints are used for all reported test metrics. The initial learning rate is set conservatively so that the batch-normalised networks converge smoothly rather than diverging early. The identical training recipe is applied to both models to preserve the fairness of the comparison.

3.9. Evaluation Metrics

The evaluation uses a comprehensive suite of metrics to give a complete picture of performance beyond aggregate accuracy, following best practice for imbalanced classification. Overall accuracy measures the fraction of correctly classified test crops and serves as the headline indicator. Per-class precision quantifies how many predictions for a class are correct, while per-class recall quantifies how many true instances of a class are found, and their harmonic

mean gives the per-class F1-score. Macro-averaged precision, recall, and F1-score weight every class equally and are therefore the fairest single summaries under imbalance. The confusion matrix, reported in both normalised and absolute form, exposes the exact structure of misclassifications between classes. One-vs-rest ROC curves and their areas under the curve summarise discriminative power independently of any single decision threshold. Training and validation curves are inspected to diagnose overfitting, and the trainable parameter count is reported as a measure of computational efficiency. Table 3 lists each metric together with its purpose.

Table 3: Evaluation metrics used in this study, with the quantity each metric captures and its role in assessing a class-imbalanced PPE compliance classifier.

Metric	Definition	Purpose
Overall accuracy	Correct / total predictions	Headline performance indicator
Precision (per class)	$TP / (TP + FP)$	Rate of false positives per class
Recall (per class)	$TP / (TP + FN)$	Rate of false negatives per class
F1-score (per class)	$2PR / (P + R)$	Balanced per-class quality
Macro F1-score	Mean of class F1	Fair summary under imbalance
Confusion matrix	Class-by-class counts	Structure of errors
AUC (one-vs-rest)	Area under ROC	Threshold-free discrimination
Train/val curves	Accuracy and loss vs epoch	Overfitting diagnosis
Parameters	Trainable weight count	Computational efficiency

3.10. Grad-CAM Interpretability Methodology

To make the classifiers auditable, Grad-CAM is applied to the compact network to visualise the image regions that drive each prediction [11]. Grad-CAM computes the gradient of the predicted class score with respect to the activations of the last convolutional layer, averages these gradients spatially to obtain channel importance weights, and forms a weighted sum of the activation maps. The resulting coarse heatmap is passed through a ReLU, normalised, and upsampled to the input resolution before being overlaid on the original crop. Warm colours indicate regions that strongly support the predicted class, while cool colours indicate regions of low influence. For a trustworthy PPE classifier, the heatmaps should concentrate on the head when a hard hat determines the decision and on the torso when a vest does. Both correctly and incorrectly classified examples are inspected so that success and failure modes can be compared directly. This interpretability step

446 transforms the quantitative results into an auditable account of model be-
 447 haviour.

448 4. Experimental Setup and Results

449 4.1. Experimental Setup

450 All experiments were implemented in Python using the TensorFlow and
 451 Keras deep-learning framework, with scikit-learn for metric computation and
 452 Matplotlib for figure generation. The person-crop extraction and label deriva-
 453 tion used the Pillow imaging library operating directly on the YOLO-format
 454 annotations of the source dataset. Both networks were trained on CPU under
 455 a fixed random seed governing the data split, weight initialisation, and aug-
 456 mentation, so that the entire experiment is reproducible. The same stratified
 457 70/15/15 split was used for both models, with 5,223 training, 1,119 valida-
 458 tion, and 1,120 test crops, and the identical augmentation and class-weighting
 459 pipeline was applied throughout. Best weights were selected by validation
 460 accuracy through checkpointing, and all reported numbers are computed on
 461 the untouched test partition. Table 4 reports the exact per-class counts in
 462 each split, confirming that stratification preserved the class proportions. This
 463 controlled setup ensures that the only material difference between the two
 464 runs is the network architecture itself.

Table 4: Per-class crop counts for the stratified 70% train / 15% validation / 15% test split. Stratification preserves the class proportions across all three partitions, so the imbalance is represented identically in training and evaluation.

Class	Total	Train	Validation	Test
PPE_Compliant	1,657	1,160	248	249
HardHat_Only	1,227	859	184	184
Vest_Only	982	688	147	147
NO_PPE	3,596	2,516	540	540
Total	7,462	5,223	1,119	1,120

465 4.2. Model Complexity Comparison

466 The two architectures differ substantially in size, which is central to the
 467 efficiency question posed in RQ5. The compact network has approximately

0.42 million trainable parameters, whereas the deeper network has approximately 2.5 million, roughly six times more. Table 5 compares the two models on layer count, parameter count, and their principal architectural characteristics. The compact model relies on single convolutions with pooling and a small head, while the deeper model uses paired strided convolutions and a larger head to gain capacity. Despite the large difference in size, both models fit comfortably within the memory and compute budget of a modest machine, which reflects the lightweight nature of the task when framed as person-crop classification. The parameter gap sets up a direct test of whether capacity translates into accuracy here.

Table 5: Architectural comparison of the two custom convolutional networks. The compact model is roughly six times smaller than the deeper model, enabling a direct test of whether added capacity improves worker-level PPE classification on this dataset.

Model	Trainable parameters	Type	Key characteristics
Compact CNN	≈ 0.42 M	From scratch	4 conv blocks, BN, max pooling, GAP head
Deeper CNN	≈ 2.5 M	From scratch	Paired strided convs, 512-filter block, larger head

4.3. Training Dynamics

The training and validation accuracy curves for the compact network are shown in Figure 5, and the corresponding curves with loss for the deeper network are shown in Figure 6. Both models begin near the chance level of one quarter and climb steadily as the convolutional features specialise to the PPE cues. The compact model converges smoothly to a validation accuracy of roughly 0.72, with training accuracy rising only slightly above validation, indicating that the regularisation successfully limited overfitting. The deeper model follows a similar trajectory but is noisier during early epochs, reflecting the harder optimisation of a larger network trained from scratch, and its loss curves show validation loss decreasing in step with training loss as learning-rate reductions take effect. Neither model exhibits the large train-validation divergence that would signal severe overfitting, which confirms that the augmentation and dropout choices were appropriate for the dataset size. The learning-rate schedule is visible as inflection points where validation accuracy jumps after a plateau. These dynamics show that both architectures train stably from random initialisation on this task.

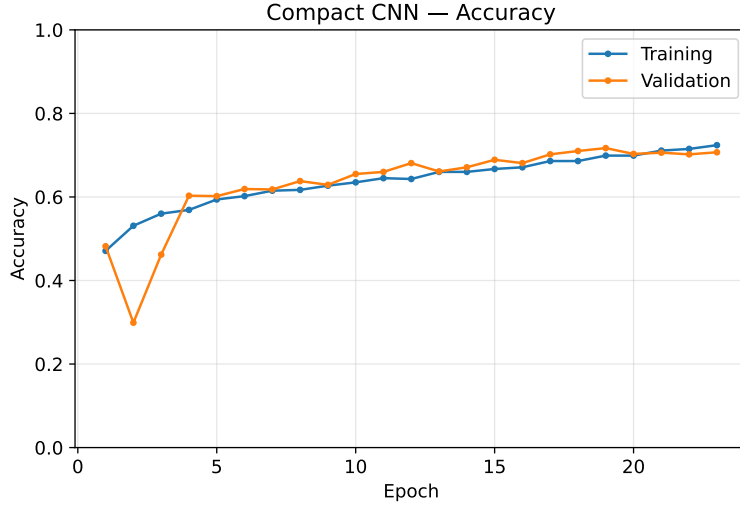


Figure 5: Training and validation accuracy of the compact custom CNN across epochs. The model climbs steadily from the chance level to a validation accuracy of roughly 0.72, with only a small gap between training and validation accuracy, indicating limited over-fitting.

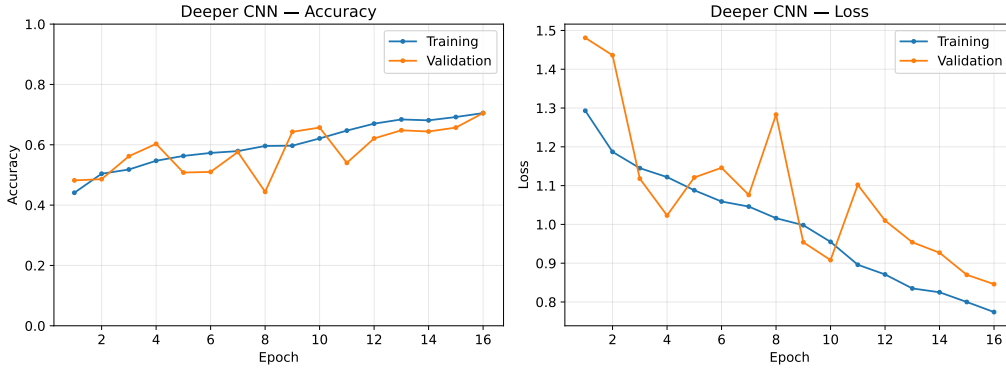


Figure 6: Training and validation accuracy (left) and loss (right) of the deeper custom CNN across epochs. The deeper network trains more noisily early on but converges to a comparable validation accuracy, with validation loss decreasing in step with training loss after learning-rate reductions.

4.4. Overall Test Performance

On the held-out test set of 1,120 crops, the compact network achieved an accuracy of 74.5 percent, a macro F1-score of 0.686, and a macro one-

vs-rest AUC of 0.902. The deeper network achieved an accuracy of 70.9 percent, a macro F1-score of 0.661, and a macro AUC of 0.888. Contrary to the common assumption that deeper is better, the compact model slightly outperformed the deeper model on every aggregate metric. This indicates that, for a dataset of roughly seven thousand crops, the additional capacity of the deeper network did not translate into better generalisation and may have made optimisation harder from scratch. Both models comfortably exceed the 25 percent chance baseline and both reach high AUC values, showing that the learned representations are genuinely discriminative. Table 6 summarises the headline comparison, and Figure 7 visualises it. The result directly answers RQ2 and RQ5: added depth did not help here, and the compact, efficient model is the stronger choice.

Table 6: Overall test-set performance of the two models on 1,120 held-out crops. The compact network slightly outperforms the deeper network on accuracy, macro F1-score, and macro AUC, despite having roughly six times fewer parameters.

Model	Accuracy	Macro F1	Macro AUC	Parameters
Compact CNN	74.5%	0.686	0.902	≈ 0.42 M
Deeper CNN	70.9%	0.661	0.888	≈ 2.5 M

4.5. Per-Class Results for the Compact CNN

Table 7 reports the full per-class precision, recall, and F1-score for the compact network. The NO_PPE class is recognised most reliably, with an F1-score of 0.82 and a high recall of 0.89, because workers with no visible equipment are the most distinctive and the most numerous. The PPE_Compliant class also scores well, with an F1-score of 0.78 and a high precision of 0.87, since the joint presence of a helmet and a vest is a strong and unambiguous signal. The Vest_Only class reaches a moderate F1-score of 0.66, reflecting that a vest is a large and colourful cue but is sometimes confused with full compliance or with no-PPE under difficult viewpoints. The hardest class is HardHat_Only, with an F1-score of just 0.48 and a low recall of 0.40, because a helmet is a small object that is easily missed at distance, from behind, or under poor contrast. This pattern confirms the expectation that partial-compliance classes, and hard-hat-only in particular, are the most challenging. The per-class results make clear that the aggregate accuracy is limited chiefly by this one difficult class rather than by uniform weakness.

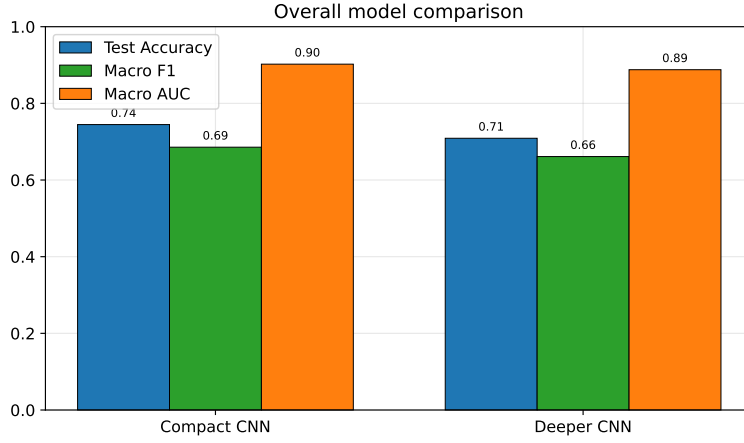


Figure 7: Overall comparison of the compact and deeper CNNs on test accuracy, macro F1-score, and macro AUC. The compact model is marginally stronger on all three aggregate metrics despite being far smaller, indicating that additional depth did not improve generalisation on this dataset.

Table 7: Per-class classification report for the compact custom CNN on the test set. The fully compliant and no-PPE classes are recognised reliably, while the hard-hat-only class is the hardest, driving down the macro F1-score.

Class	Precision	Recall	F1-score	Support
PPE_Compliant	0.87	0.71	0.78	249
HardHat_Only	0.58	0.40	0.48	184
Vest_Only	0.64	0.69	0.66	147
NO_PPE	0.76	0.89	0.82	540
Accuracy		0.745		1,120
Macro average	0.71	0.67	0.69	1,120
Weighted average	0.74	0.74	0.73	1,120

4.6. Per-Class Results for the Deeper CNN

Table 8 reports the per-class results for the deeper network. Interestingly, the deeper model spreads its performance more evenly across classes, improving the hard HardHat_Only class to an F1-score of 0.53 while giving up ground on the PPE_Compliant and NO_PPE classes. Its NO_PPE F1-score falls to 0.78 and its PPE_Compliant F1-score to 0.77, so despite the gain on the hardest class its macro F1-score of 0.661 remains below the

compact model’s 0.686. This trade-off suggests that the extra capacity re-distributes errors rather than eliminating them, helping the minority class slightly at the expense of the easier classes. The Vest_Only class remains moderate at 0.57, again reflecting occasional confusion with adjacent classes. The overall picture is of two models with similar strengths and weaknesses, differing mainly in how they balance the easy and hard classes. This reinforces the conclusion that depth alone does not resolve the fundamental difficulty of the partial-compliance classes.

Table 8: Per-class classification report for the deeper custom CNN on the test set. The deeper model improves the hard-hat-only class relative to the compact model but loses ground on the easier classes, leaving its macro F1-score slightly lower overall.

Class	Precision	Recall	F1-score	Support
PPE_Compliant	0.76	0.78	0.77	249
HardHat_Only	0.53	0.53	0.53	184
Vest_Only	0.57	0.57	0.57	147
NO_PPE	0.78	0.78	0.78	540
Accuracy		0.709		1,120
Macro average	0.66	0.66	0.66	1,120
Weighted average	0.71	0.71	0.71	1,120

4.7. Per-Class F1 Comparison

Figure 8 compares the per-class F1-scores of the two models directly. The compact model leads on the PPE_Compliant and NO_PPE classes, while the deeper model leads on the HardHat_Only and Vest_Only classes. This crossover pattern makes explicit the trade-off described above: the compact model is better on the easy, frequent classes, and the deeper model is marginally better on the harder, rarer ones. Because the frequent classes carry more weight in overall accuracy, the compact model wins on the aggregate metrics. The figure also shows that both models agree on the difficulty ordering of the classes, with hard-hat-only lowest and no-PPE highest for the compact model. This consistency across two independently trained architectures strengthens confidence that the difficulty ordering reflects the data rather than a training artefact. The comparison directly informs the answer to RQ4 about which class is hardest.

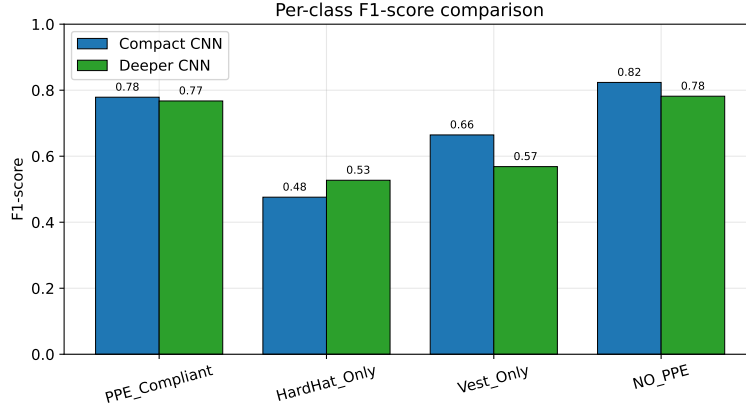


Figure 8: Per-class F1-score comparison between the compact and deeper CNNs. The compact model is stronger on the frequent PPE_Compliant and NO_PPE classes, while the deeper model is slightly stronger on the harder HardHat_Only and Vest_Only classes, illustrating a redistribution of errors rather than a uniform improvement.

556 4.8. Confusion Matrix Analysis

557 Figure 9 shows the normalised and absolute confusion matrices for the
558 compact network, and Figure 10 shows them for the deeper network. For
559 the compact model, the dominant errors are HardHat_Only crops misclas-
560 sified as NO_PPE, which occurs when the helmet is too small or indistin-
561 ct to register, so the worker appears unequipped. There is also notable con-
562 fusion between PPE_Compliant and the two partial classes when one item is
563 occluded or seen at an angle. The NO_PPE class is rarely confused with
564 anything else, consistent with its high recall. The deeper model’s confu-
565 sion matrix shows the same principal error direction but with the mass distrib-
566 uted slightly differently, recovering more HardHat_Only cases at the cost of a few
567 more errors on the easier classes. In both matrices the errors are structured
568 and interpretable rather than random, concentrating between classes that
569 share visual appearance or differ only by a small, easily missed item. This
570 structure points directly to occlusion, distance, and viewpoint as the root
571 causes, which the Grad-CAM analysis then confirms.

572 4.9. ROC and AUC Analysis

573 Figure 11 presents the one-vs-rest ROC curves for both models and all
574 four classes. Every curve lies well above the diagonal chance line, and
575 the macro AUC values of 0.902 for the compact model and 0.888 for the

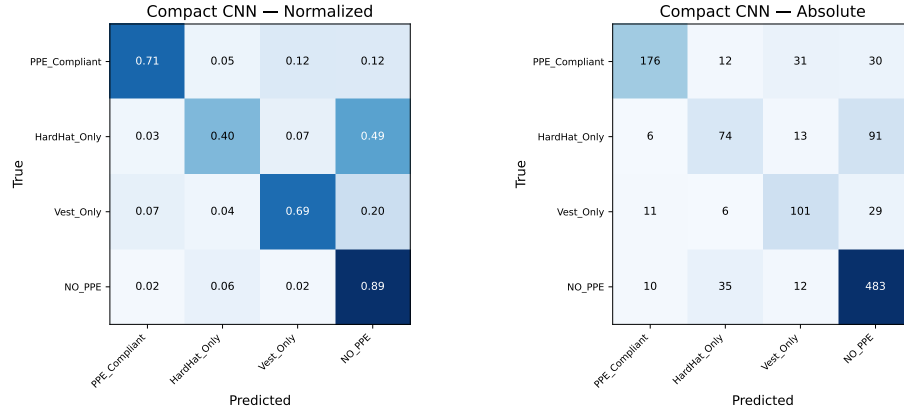


Figure 9: Normalised (left) and absolute (right) confusion matrices for the compact custom CNN on the test set. The largest error is HardHat_Only crops predicted as NO_PPE, reflecting missed small or distant helmets, while the NO_PPE class is rarely confused with any other.

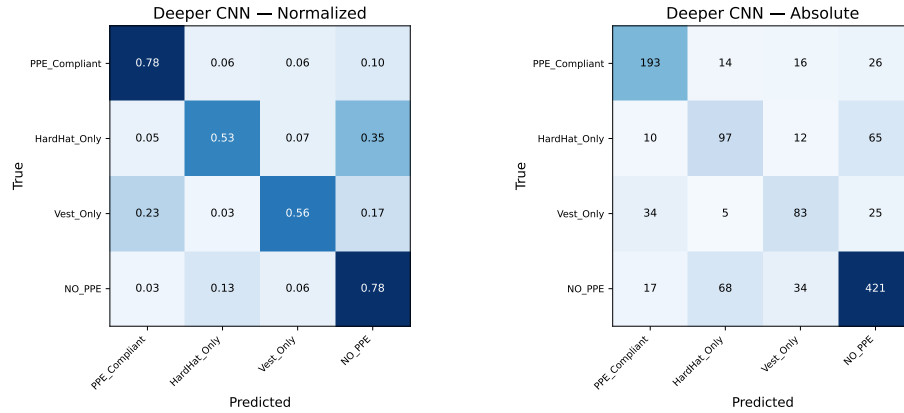


Figure 10: Normalised (left) and absolute (right) confusion matrices for the deeper custom CNN on the test set. The error structure mirrors that of the compact model but redistributes some mass, recovering additional hard-hat-only cases at the expense of a few errors on the easier classes.

576 deeper model confirm strong threshold-independent discrimination. The
577 NO_PPE and PPE_Compliant classes show the highest AUCs, consistent
578 with their high F1-scores, while the partial-compliance classes have lower
579 but still clearly useful AUCs. The ROC analysis shows that even for the
580 hard HardHat_Only class the models rank positive instances above nega-

581 tives far better than chance, which means a tuned decision threshold could
 582 trade recall for precision as an application requires. The closeness of the two
 583 models' curves reiterates that depth made little difference to overall discrim-
 584 ination. Because AUC is insensitive to the class imbalance, these results give
 585 a reassuring view of model quality that complements the accuracy figures.
 586 The ROC evidence supports the conclusion that the classifiers are genuinely
 587 discriminative across all classes, not merely accurate on the majority class.

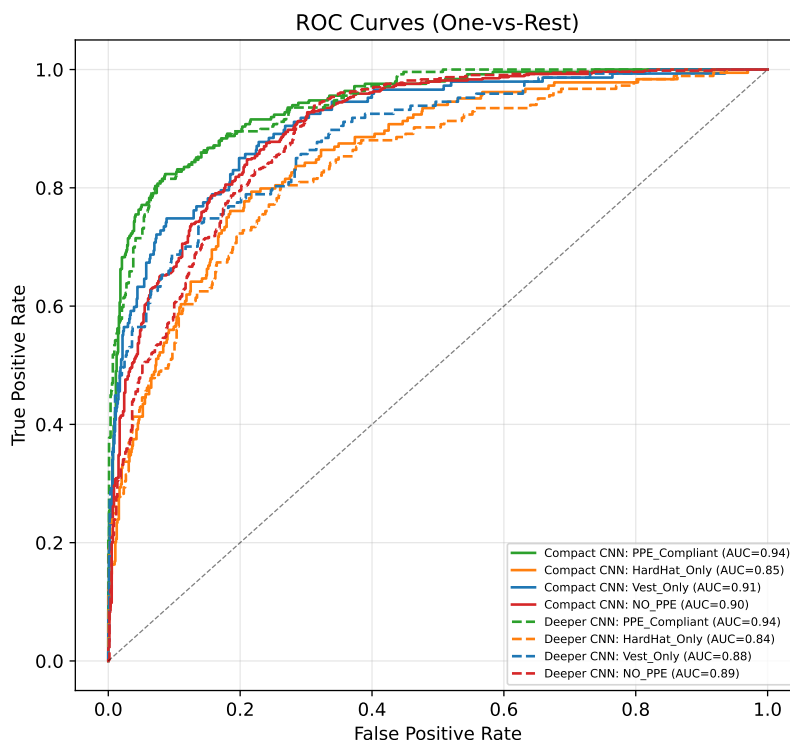


Figure 11: One-vs-rest ROC curves for the compact (solid) and deeper (dashed) CNNs across the four PPE compliance classes. All curves lie well above the chance diagonal; the compact model attains a macro AUC of 0.902 and the deeper model 0.888, confirming strong threshold-independent discrimination even for the harder classes.

588 4.10. Grad-CAM Interpretability Results

589 Figure 12 shows Grad-CAM overlays for representative correctly and in-
 590 correctly classified crops from the compact model. For correctly classified
 591 compliant and hard-hat cases, the heatmaps concentrate on the head region,

confirming that the model uses the helmet as the intended cue. For vest cases, the attention shifts to the torso, again matching the location of the discriminative equipment. This alignment between model attention and the true PPE location is strong evidence that the classifier has learned semantically meaningful features rather than background shortcuts. In the misclassified examples, the heatmaps reveal the failure mechanism: attention drifts to background objects, to bright clothing that resembles a vest, or spreads diffusely when the helmet is small or occluded. These visual explanations directly corroborate the confusion-matrix finding that occlusion, distance, and colour similarity drive the errors. By exposing both correct reasoning and failure modes, the Grad-CAM analysis makes the classifier auditable and answers the interpretability part of RQ4.

5. Discussion

5.1. Answering the Research Questions

The results allow each research question to be answered directly and honestly. For RQ1, a convolutional network trained from scratch does achieve a useful and balanced classification of the four compliance states, reaching 74.5 percent accuracy and a macro AUC above 0.90, which is well above chance and discriminative across all classes, though short of the very high accuracies reported by pre-trained detection systems. For RQ2, the deeper architecture did not outperform the compact one under identical conditions; the compact model was marginally better on every aggregate metric, showing that added depth from scratch on this dataset size did not improve generalisation. For RQ3, conservative augmentation of flipping, small rotation, and zoom combined with softened class weighting and dropout proved effective at controlling overfitting, as evidenced by the small train-validation gap, whereas heavier augmentation was found to hinder convergence. For RQ4, the `HardHat_Only` class is unambiguously the hardest, and Grad-CAM and the confusion matrices show that missed small or distant helmets, occlusion, and colour confusion are the causes. For RQ5, the compact model, with roughly 0.42 million parameters, matches or exceeds the deeper model while being far cheaper, so a lightweight model is not only sufficient but preferable for real-time edge deployment.

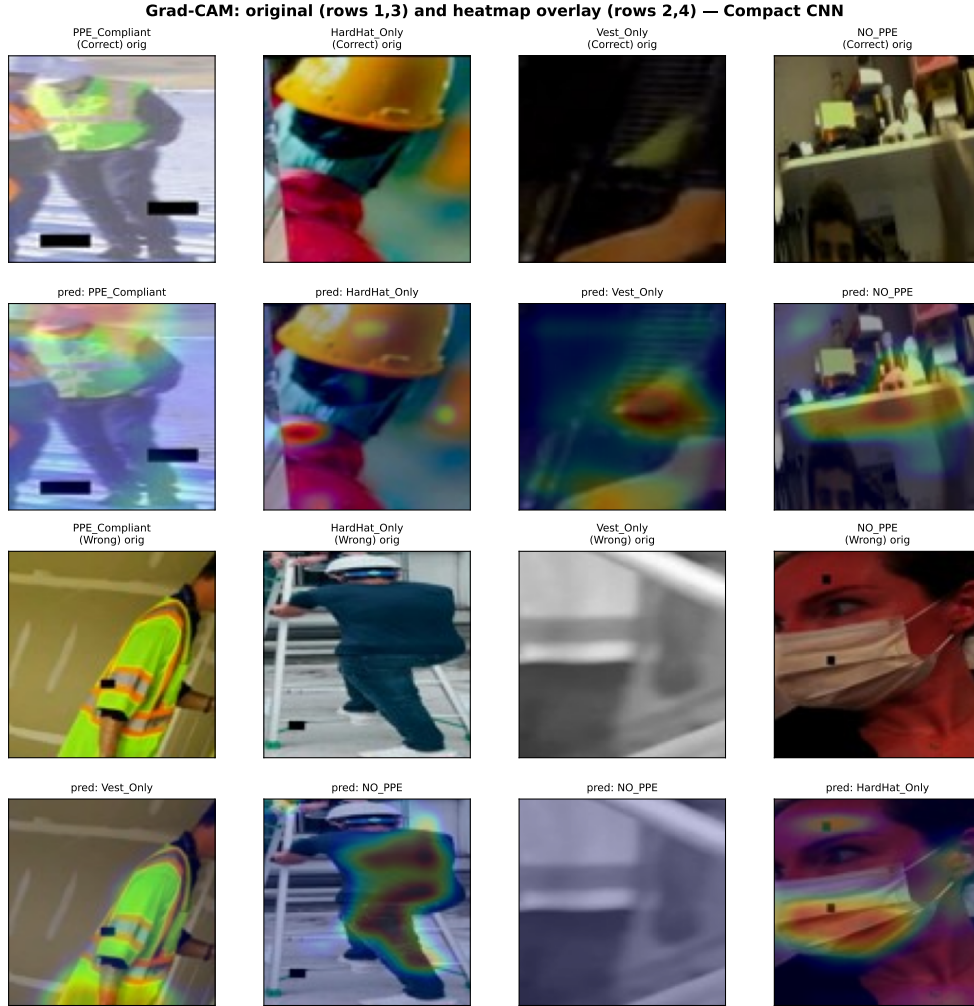


Figure 12: Grad-CAM overlays for the compact CNN on representative correctly classified (rows 1–2) and misclassified (rows 3–4) crops across the four classes. Warm regions indicate high influence on the prediction. Correct cases attend to the head for helmets and the torso for vests, while misclassifications show attention drifting to background or ambiguous colour cues.

5.2. Interpretation and Comparison with Literature

The finding that a compact from-scratch model is competitive with, and here better than, a deeper one echoes the broader lesson that capacity must be matched to data size, and that very deep networks can struggle to gen-

629 eralise on modest datasets without pre-training [4, 22]. The absolute accu-
630 racies are lower than the high nineties frequently reported in PPE detection
631 papers, but this is expected and honest, because those systems typically use
632 pre-trained backbones, larger inputs, and a detection framing that benefits
633 from precise localisation [1, 6]. The present study deliberately measures the
634 raw difficulty of worker-level classification from scratch, which the literature
635 rarely does, and the results quantify how much of the task’s difficulty is con-
636 centrated in the partial-compliance classes. The Grad-CAM evidence that
637 the model attends to the head and torso is consistent with the interpretabil-
638 ity expectations expressed across the explainable-AI literature [11, 10]. The
639 structured confusions between visually adjacent classes also mirror obser-
640 vations in prior safety-recognition studies where similar items are conflated
641 under difficult imaging conditions [17]. Overall the results are coherent with,
642 and add nuance to, the existing body of work. Their value lies in trans-
643 parency rather than in setting a new accuracy record.

644 5.3. *Practical Implications*

645 For a deployed safety system, the results carry several practical mes-
646 sages. A lightweight person-crop classifier can provide a genuinely useful
647 first-pass compliance signal at very low computational cost, which is attrac-
648 tive for on-site edge hardware and for processing many camera streams in
649 parallel. Because the NO_PPE and PPE_Compliant classes are recognised
650 reliably, the system is well suited to flagging the clearest and most safety-
651 critical cases, namely workers with no equipment at all. The weakness on
652 the HardHat_Only class means that a real deployment should treat partial-
653 compliance predictions with appropriate caution, for example by escalating
654 uncertain cases to a human reviewer. Coupling the classifier with an up-
655 stream person detector, as in the two-stage designs common in the litera-
656 ture, would allow it to run directly on full camera frames. The threshold-free
657 strength shown by the ROC analysis means operators could tune the decision
658 threshold to prioritise recall of non-compliance, which is the safety-relevant
659 error to avoid. These implications position the model as a useful component
660 rather than a complete standalone safety solution.

661 5.4. *Limitations*

662 This study has several limitations that should temper its conclusions.
663 First, no pre-trained backbones were used, because ImageNet weights were
664 unavailable in the training environment, so the reported accuracies represent

665 a from-scratch lower bound rather than the best achievable with transfer
 666 learning. Second, the input resolution was limited to 96×96 pixels for com-
 667 putational reasons, which discards fine detail that is especially important
 668 for recognising small helmets, and higher resolution would likely improve
 669 the hardest class. Third, the compliance labels were derived automatically
 670 from bounding-box annotations using a containment heuristic, which can in-
 671 troduce occasional labelling noise when boxes overlap ambiguously. Fourth,
 672 the dataset, though realistic, is moderate in size and drawn from a limited
 673 range of sites, so the models may not generalise to very different environ-
 674 ments, camera types, or PPE styles. Fifth, the study evaluates only two
 675 custom architectures and does not include the pre-trained MobileNetV2 or
 676 ResNet comparisons that would place the results against standard baselines.
 677 Sixth, the evaluation is offline on still crops and does not measure real-time
 678 throughput or temporal consistency on video. These limitations are stated
 679 plainly so that the results are not over-interpreted.

680 5.5. Future Work

681 Several concrete directions would build on this study. The most imme-
 682 diate is to add transfer-learning baselines using MobileNetV2 and ResNet50
 683 with pre-trained weights once network access permits, which the literature
 684 suggests would substantially raise accuracy, especially on the hard classes
 685 [9, 8]. Increasing the input resolution and, where possible, tightening the
 686 person crops around the upper body would give the model more of the fine
 687 detail needed to detect small helmets. Integrating the classifier with a ded-
 688 icated person detector into a single end-to-end pipeline would allow direct
 689 operation on full construction-scene frames and enable evaluation of real-time
 690 performance [16]. Expanding and diversifying the training data across more
 691 sites, weather conditions, and PPE variants would improve robustness and
 692 external validity. Targeted techniques for the minority classes, such as focal
 693 loss, oversampling, or class-specific augmentation, could directly attack the
 694 hard-hat-only weakness. Finally, extending the interpretability analysis with
 695 quantitative faithfulness metrics and testing temporal smoothing on video
 696 would strengthen both trust and deployability. Together these steps form a
 697 clear path from the honest baseline established here to a production-grade
 698 safety system.

699 6. Conclusion

700 This study developed and rigorously evaluated an explainable convolu-
701 tional neural network framework for worker-level PPE compliance classifica-
702 tion on construction-site imagery. By extracting 7,462 person-centred crops
703 from a public detection dataset and deriving four mutually exclusive compli-
704 ance labels, it reframed a scene-level detection problem as a tractable and
705 directly actionable classification task. Two custom architectures of differ-
706 ing depth were trained from scratch under strictly identical conditions, and
707 the compact 0.42-million-parameter network reached 74.5 percent test accu-
708 racy, a macro F1-score of 0.686, and a macro AUC of 0.902, slightly outper-
709 forming the six-times-larger deeper network. A complete per-class analysis
710 showed that the fully compliant and no-PPE classes are recognised reliably
711 while the hard-hat-only class is by far the most difficult, and Grad-CAM
712 confirmed that the model bases its correct decisions on the head and torso
713 regions where PPE is worn. The principal conclusions are that a small effi-
714 cient model is sufficient and even preferable for this task, that added depth
715 does not automatically help on a modest dataset trained from scratch, and
716 that the remaining difficulty is concentrated in partial-compliance cases af-
717 fected by occlusion, distance, and colour confusion. The main contribution
718 is an honest, reproducible, and interpretable baseline that quantifies exactly
719 where the task is easy and where it is hard, rather than a single headline
720 number obtained with hidden advantages. With the addition of pre-trained
721 backbones, higher-resolution inputs, and integration with a person detector,
722 this framework offers a clear and credible path toward a deployable real-time
723 construction-safety monitoring system.

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